CCGA Communication Gateway Instructions for GE Lufkin

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Safety Symbol Legend



Warning

Indicates a procedure, condition, or statement that, if not strictly observed, could result in personal injury or death.



Caution

Indicates a procedure, condition, or statement that, if not strictly observed, could result in damage to or destruction of equipment.



Attention

Indicates a procedure, condition, or statement that should be strictly followed to improve these applications.

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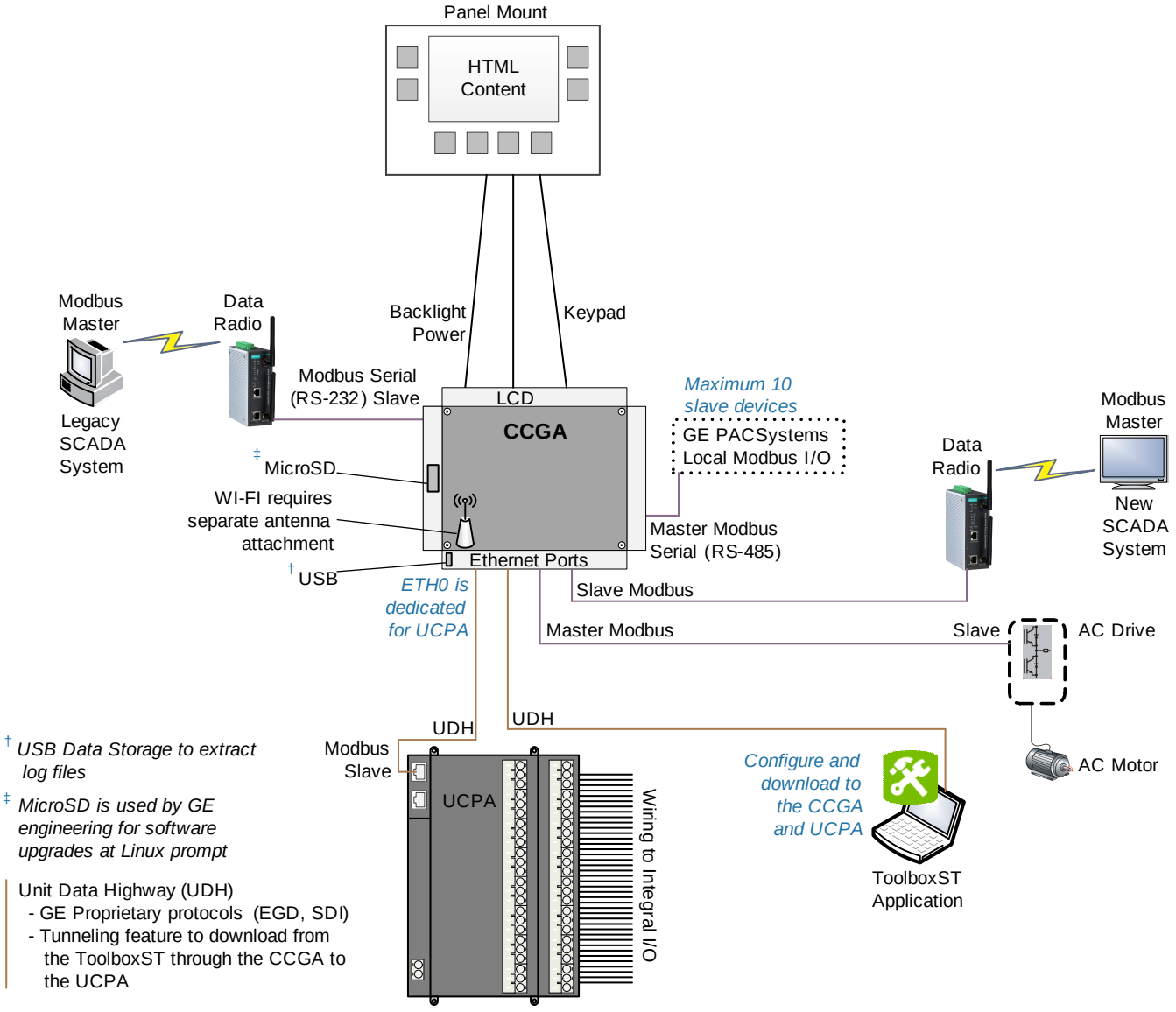
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Related Documents

Title	Number
ToolboxST* User Guide for CCGA Communication Gateway	GEI-100848
ToolboxST* User Guide for Mark* VIe Control	GEH-6700
CCGA Communication Gateway Instructions for Safe Use	GFK-2953
CCGA Communication Gateway Instructions for GE Lufkin	GEK-SA-1045 (this document)
Mark* VIe UCPA Controller Instructions for Safe Use	GFK-2951
Mark* VIe UCPA Controller with Integral I/O Instruction Guide	GEI-100719
WEXP Board for the UCPA Controller Instructions for GE Lufkin	GEK-SA-1047

1 CCGA Communication Gateway Module

The Communication Gateway Module (CCGA) is a Remote Terminal Unit (RTU) that runs a Linux operating system. This module mounts to the door of a GE Lufkin box. The CCGA communicates through a dedicated Ethernet port (ENET0) to a Mark VIe UCPA controller (also mounted in the GE Lufkin box). The CCGA provides communication options for many types of Modbus devices (including PACSystems* local I/O) as displayed in the following figure.



CCGA Available Communications

1.1 Features

- Single module
- No fan
- Built in WI-FI option
- GSM/CDMA connectivity option
- RS-485, RS-232 communication ports
- USB and Micro SD slots
- A low cost RTU
- Firmware and application provides a Modbus® Register Map and web interface to communicate between Mark* VIe controller (UCPA), SCADA, and HMI screens
- Well manager HTML5 screens with data connection
- ToolboxST* configurator for downloading to the CCGA



CCGA Communication Gateway

1.2 Specifications

Item	CCGA Specification
Processor	TI AM3354 32-bit ARM processor, 600MHz
Memory	512MB DDR3 RAM
	1GB on-board flash
Ambient Temperature, Humidity	-40°C to 70°C, 5 TO 95% non-condensing
Battery backup	Battery backup for RTC, 6 yrs
USB Port	Used to retrieve event logs, for use with Lufkin Modbus tables
SD Card Slot	One MicroSD card slot for firmware upgrade, configuration downloading and uploading event records
Ethernet Ports	Four Ethernet ports, 10/100 Mbps (ENET0 — ENET3) 3–port integral Ethernet switch (ENET1 — ENET3)

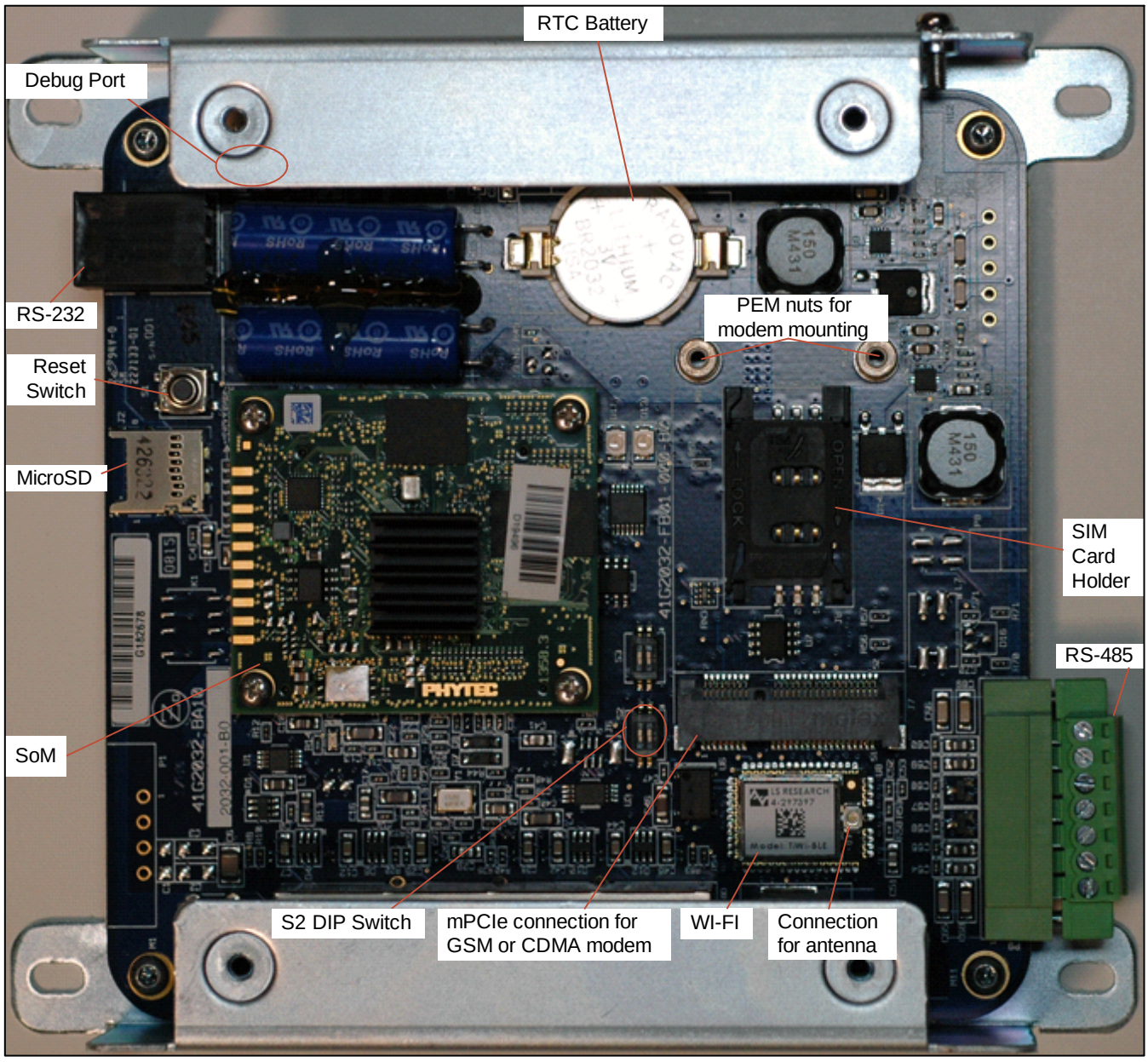
Item	CCGA Specification
RS-232	One RS–232 Serial Port (RJ–12) Maximum cable distance is 15 m Maximum communication rate is 115.2 kbps
RS-485	One RS–485 Serial Port (pluggable screw terminal)
	Maximum cable distance: 305 m (1000 ft)
	Maximum communication rate: 115.2 kbps
Serial Protocols	Modbus master/slave RTU
Ethernet protocols	Modbus TCP client/server, EGD
Wireless	H2A version has WI-FI (requires connection of antenna) Optional 3G, GSM (with installation of SIM card)
Gateway configuration	ToolboxST application
Operating System	Embedded Linux
LCD Display Interface	Supports Lufkin 640x480 TFT Display (AUO Make p/n# G065VN01 V2)
Keypad Interface	To GE Lufkin custom keypad
Operating voltage	9 to 16 V dc, nominal 12 V dc
Power consumption	12.2 Watts nominal, estimate
3-pin power plug *included with the CCGA part kit*	Wire sizes: 22 to 14 AWG Screw torque: 2 in-lb Temperature rating for copper wire: 80 °C Wiring to power input terminals shall be limited to 30 meters in length
Housing dimensions	6.15 x 5.8 x 1.25 Inches (156.21 x 147.32 x 31.75 mm)
Mounting	Base mounted

1.3 Internal Components

Refer to the section, Debug Port for more information.

The CCGA module is made up of the following components:

- 41G2032BA10-000-A0 or 41G2032BA10-001-A0 carrier board
- 42G4000-0139 system on module (SoM)



CCGA Internal Components

Note Use a TORX T8 screwdriver to remove the CCGA cover.

1.3.1 System on Module (SoM)

The 42G4000-0139 system on module (SoM) is a phyCORE-AM335x, belonging to PHYTEC's phyCORE System on Module family. The phyCORE-AM335x is a subminiature (44 mm x 50 mm) insert-ready System on Module populated with the Texas Instruments AM335x microcontroller. Its universal design enables its insertion in a wide range of embedded applications. All controller signals and ports extend from the controller to high-density pitch (0.5 mm) connectors aligning two sides of the board, allowing it to be plugged like a "big chip" into a target application.

The phyCORE-AM335x offers the following features:

- Insert-ready, sub-miniature (44 mm x 50 mm) System on Module (SOM) subassembly in low EMI design, achieved through advanced SMD technology
- Populated with the Texas Instruments AM335x microcontroller (ZCZ 324-pin PBGA package)
- Max. 600 MHz core clock frequency
- Boot from NAND Flash or SPI Flash

1.4 CCGA Module Versions

The IS420CCGAH1A includes the 41G2032-BA10-001 carrier board and 42G4000-0139 SoM. It provides the following features.

- 4 Ethernet ports
  - ENET0 dedicated to communicate with UCPA for UDH and TCP Modbus
  - ENET1 to ENET3 with integral 3-port switch for TCP Modbus Master/Slave protocol and UDH GE protocols
- RS-485 supporting Modbus protocol
- RS-232 port for Data Radio configuration
- USB port
- MicroSD interface for firmware upgrades using a Linux prompt
- LCD & Backlight display Interface supports 640x480 TFT Display
- Keypad Connector
- mPCIe connector for cellular internet GSM or 3G CDMA supported

The IS420CCGAH2A includes the 41G2032-BA10-001 carrier board and 42G4000-0139 SoM. It provides all of the features of the CCGAH1A and additionally has built in WI-FI to allow the viewing of web pages over a wireless network.

1.5 Installation

The CCGA is contained in a single module that mounts directly to the panel door. The CCGA is base mounted and has dedicated shield tie. The top and bottom part of the package is powder coated sheet metal.

ENET0 is the dedicated port for communication to the UCPA controller. Files that are located on the computer that runs the ToolboxST application may need to be modified or installed separately in support of the HTML5 web pages, EGD data maps, and Modbus data maps. Refer to the *ToolboxST User Guide for CCGA Communication Gateway (GEI-100848)* for more information, including instructions for configuration and downloading to the CCGA.

Connection of an external antenna using a micro BNC connector is required for WI-FI communications (CCGAH2A version only). Connection of a Telit cell phone modem and sim card provides cellular internet capability.

1.5.1 Power Requirements

The CCGA runs on a nominal 12 V dc power supply (9 to 16 max). The 12 V dc power input is intended to be powered from a supply with Class II protection level. Wiring to the CCGA power input shall be no more than 30 meters in length.

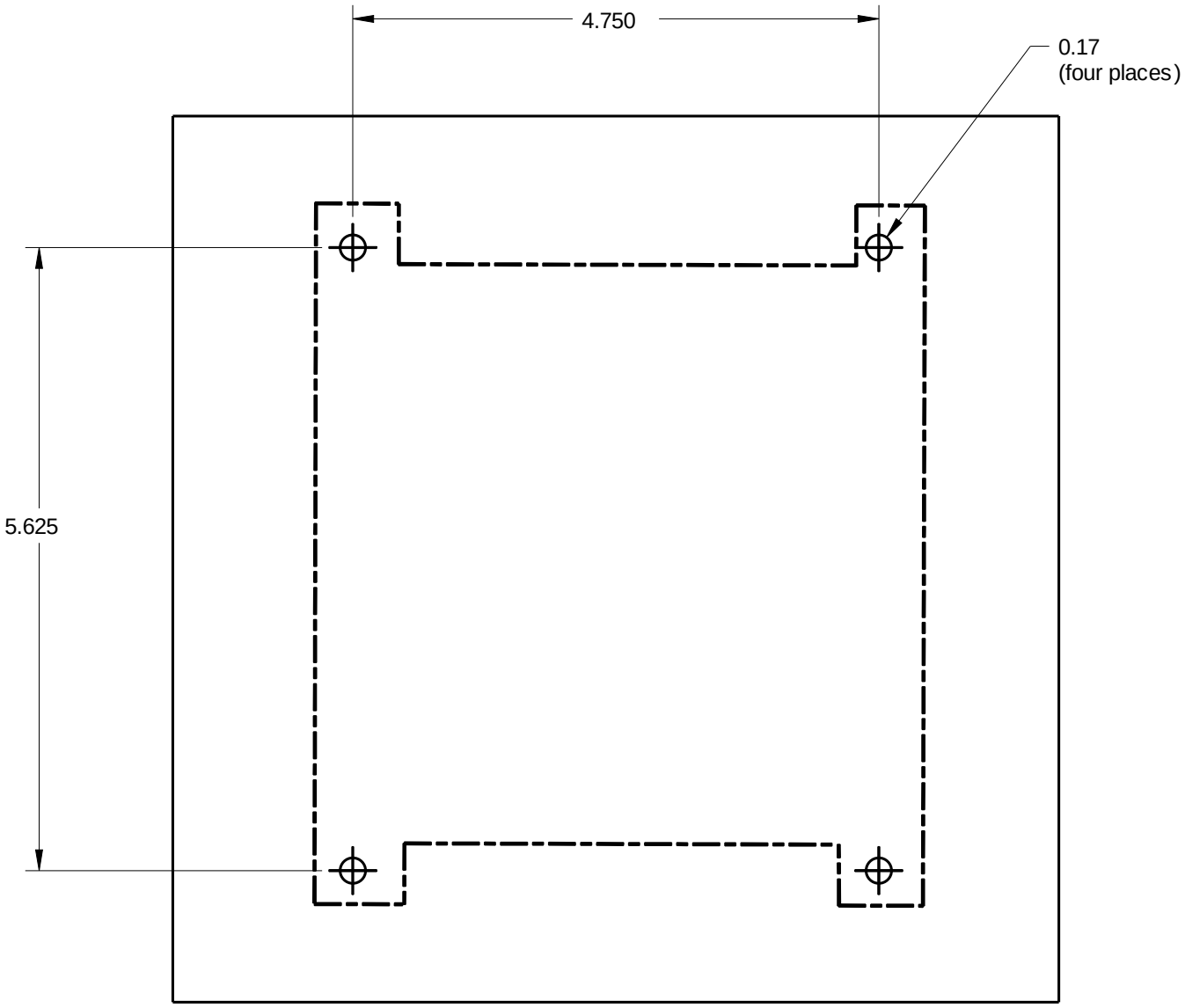


Caution

Reversing input power polarity might cause damage to the CCGA. More than 16 V dc power to the CCGA can cause damage to the interfaced LCD display (AUO : G065VN01_V2).

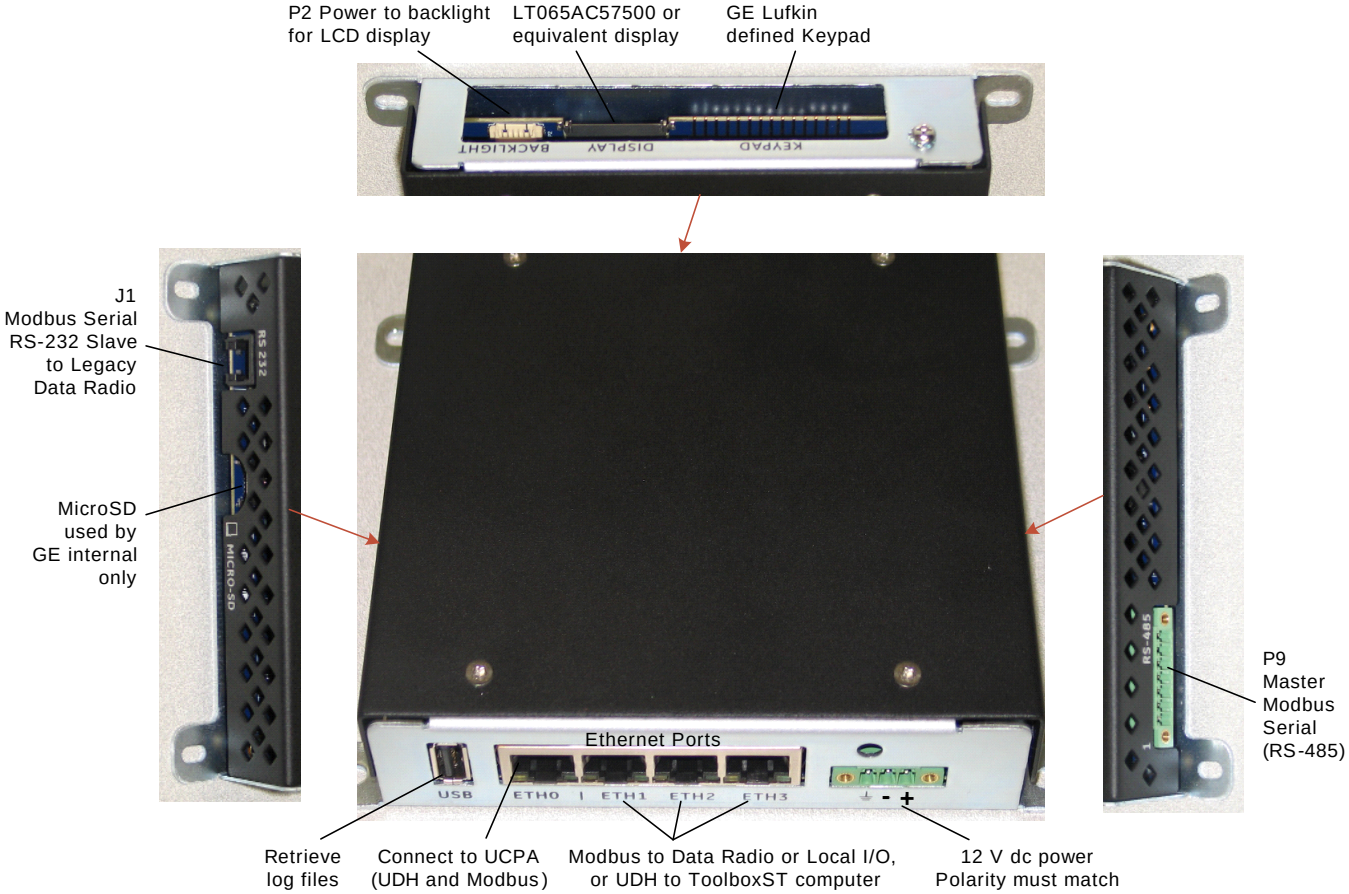
1.5.2 Mounting

The following figure displays the mounting dimensions of the CCGA module in inches.



1.5.3 Interface Details

The following figure displays all four sides of the CCGA module with details for connections



1.5.4 Pin Definitions

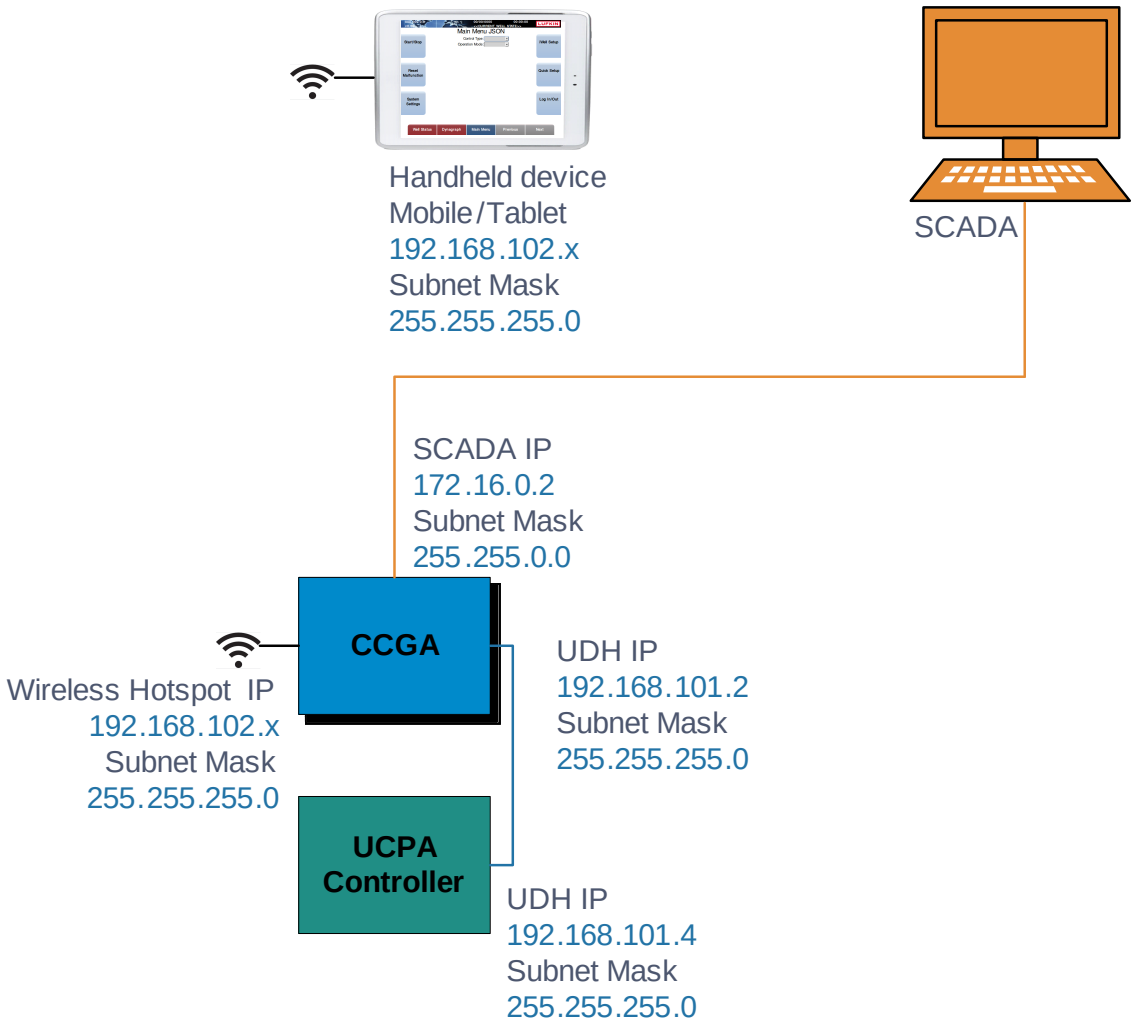
Pin Definitions

Connector	Function	Pins
J1	RS-232	1 NC
		2 TxD
		3 Rx
		4 GND
		5 RTS
		6 CTS
P2	Back Light Power	1 Vin
		2 Vin
		3 GND
		4 GND
		5 LED ON/OFF
		6 PWM DIM
		7 GND
		8 GND
P9	RS-485	1 Shield
		2 Shield
		3 GND
		4 Rx+
		5 Rx-
		6 Tx+
		7 Tx-
J4	Debug Serial Port (RS-232)	1 NC
		2 Tx
		3 Rx
		4 GND

1.5.5 UDH Network IP Address

Refer to the ToolboxST User Guide for CCGA Communication Gateway (GEI-100848) for more detailed instructions.

The CCGA module comes from the factory with a default IP address for ENET0 (connection to the UCPA controller) and for ENET1 (typically connected to the ToolboxST computer). Use these IP addresses in the initial ToolboxST application to go Online with the components, and then change the IP Address as required (then download the new IP addresses if needed).



GE Lufkin Default IP Addresses

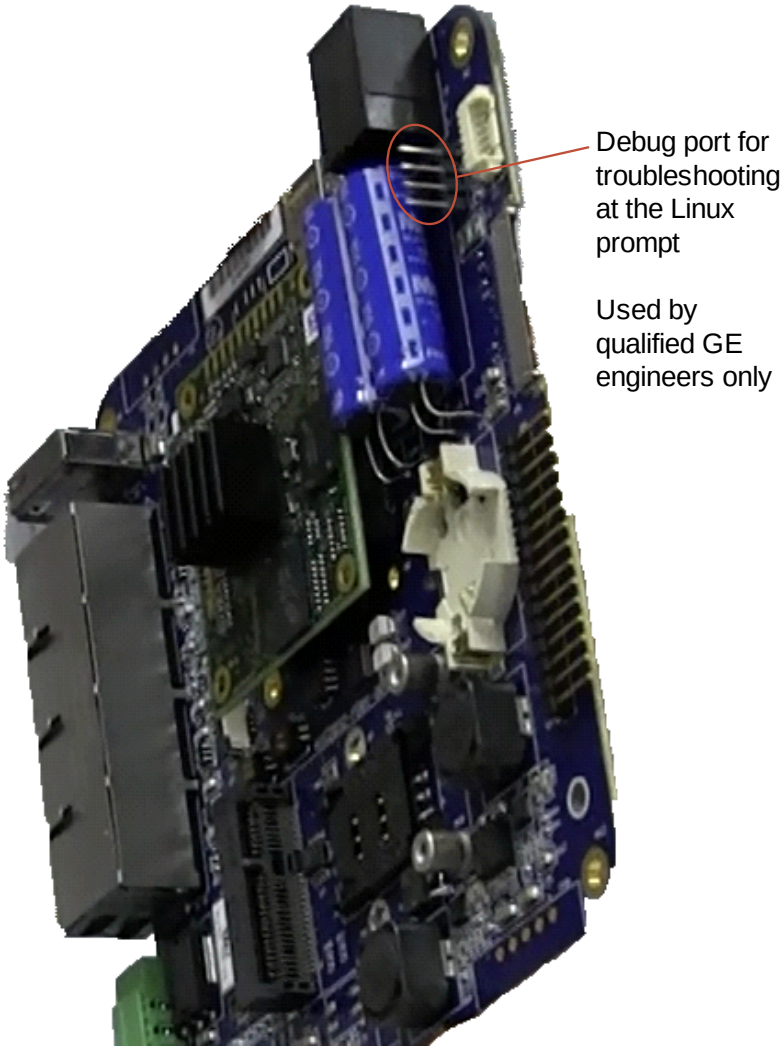
1.6 Modbus and UDH EGD XML Files

Refer to the ToolboxST User Guide for CCGA Communiation Gateway (GEI-100848) for more information.

The CCGA firmware runs both Modbus and UDH EGD data maps that can be modified for specific GE Lufkin applications by editing xml files, which are located on the ToolboxST computer. Then, these files are dowloaded from the ToolboxST application to the CCGA module.

1.7 Debug Port

Only a trained GE engineer shall use this port. The default serial communication setting is 115200 8-n-1. The debug port is the only interface that can display boot logs or run boot-level system commands. Use GE part # 116T7500P0001 serial cable for connection to this port.



1.8 LEDs

There are two LEDs (Yellow/Green) for each Ethernet connector (ENET0 to ENET3), which are embedded in the RJ-45 connectors. The green LED indicates an Ethernet connection has been established. The yellow LED indicates packet traffic.

1.9 Log Files

For troubleshooting, the CCGA has a USB port that is used to extract log files. The purpose of *LogMessage.txt* file is to record *Well Manager* firmware runtime status and firmware version related messages. If the file size is more than 5 MB, then *LogMessage.txt* is backed up as *LogMessages_Old.txt* and a new file *LogMessages.txt* is created. These files are placed in */home/* folder. The Following are details of the Log file.

At power on, an iWelllMgr startup message is logged and followed by Product Version, Baseload Version, RamFS Version, Well Controller IP, Status of subsystem initialization, and Status of connection between *Well Manager*(CCGA) and *Well Controller* (UCPA). Below are the messages logged in LogMessage.txt at startup.

```
iWellMgr startup ...................
Product Version 00.00.00 - 00000
Baseload Version 00.00.00 - 00000
RamFS Version 00.00.00 - 00000
0000000018.0146385540 Well controller IP - 192.168.101.115
0000000018.0651268352 All Initializations complete
0000000023.0545585978 MMS - Connection established with well
controller
```

Messages are logged using this format: "<timestamp> <log message>"

The following is an example:
`0000000018.0146385540 Well controller IP - 192.168.101.115`
"0000000018.0146385540" is a timestamp in format **seconds.nanoseconds**.
"Well controller IP - 192.168.101.115" is a log message.

Startup events, and events that occur while running Well Manager firmware are stored in Modbus register address 30603. A corresponding event timestamp is stored in 30803. Specific Event Id ranges are given to individual subsystems. Users can check the type of event occurred from the Modbus register 30603. The details of the events are as follows.

Event Id	Description	Comments
Modbus Master (500 – 999)		
500	Modbus connection with Well Controller failed	
501	Modbus connection with Well Controller resumed	
502-999	Reserved	
Modbus slave (1000 – 1499)		
1000	Modbus TCP Client connected	
1001	Modbus TCP Client disconnected	
1002 - 1499	Reserved	
EGD (1500 – 1999)		
1500	EGD other errors	
1501	Page1_FrameRate Exchange communication timed out	1. If any exchange times out, the 'Dynamic EGD communication' fail bit will be set in Modbus register 10674
1502	Page1_FrameRate Exchange communication recovered	2. When all exchanges recover from a communication failure, the 'Dynamic EGD communication' fail bit will be cleared in Modbus register 10674

Event Id	Description	Comments
1503	Page2_500ms Exchange communication timed out	3. Whenever an exchange times out, the 'Latch EGD communication' fail bit will be set in Modbus register 10673. However 'Latch EGD communication' fail bit will not be cleared even when all exchange communication resumes. This bit will be cleared only when user issues a “clear all alarms” command by setting 0x00002 register.
1504	Page2_500ms Exchange communication recovered	
1504 - 1999	Reserved	
XML (2000 – 2499)		
2000	ccgateway-modbus-outputs.xml format error	Error while parsing the file
2001	ccgateway-modbus-outputs.xml file operation error	Error while opening the file, could be file missing
2002	ccgateway-egd-outputs.xml format error	Error while parsing the file
2003	ccgateway-egd-outputs.xml file operation error	Error while opening the file, could be file missing
2004 - 2499	Reserved	
Alarms (2600 – 2699)		
2600	Alarms log corruption	
2601	Invalid alarm track input state register address loaded	
2602	Invalid alarm tracking mode	
2603	Invalid alarm tracking state	
2604 - 2699	Reserved	
Register Log (2700 – 2799)		
2700	Invalid Register Log descriptor	
2701	Invalid Register Log data type	
2702	Invalid register Log Modbus data type	
2703	Invalid register Log slave Modbus datatype	
2704	Invalid register Log slave address	
2705	Invalid register log type	
2706	Invalid analog input number	
2707	Invalid accumulator input number	
2708	Invalid register log state	
2709	Invalid register log rate	
2710	Invalid Modbus register address	
2711	Invalid Modbus slave register address	
2712 - 2799	Reserved	
Modbus Slave Communication (2800 – 2999)		
2800	Thread create error	
2801	Invalid state control	Modbus Master device enable/disable state

Event Id	Description	Comments
2802	Invalid baud rate	
2803	Invalid slave address	
2804	Invalid data bits	
2805	Invalid stop bits	
2806	Invalid parity	
2807	Invalid timeout	
2808	Invalid retries	
2809	Invalid device type	
2810	Modbus context error	
2811	Modbus connection error	
2812	Modbus mode error	
2813 - 2899	Reserved	
2900	VFD Thread Create Error	
2901	Invalid VFD Baud Rate	
2902	Invalid VFD Slave ID	
2903	Invalid VFD data bits	
2904	Invalid VFD stop bits	
2905	Invalid VFD parity	
2906	Invalid VFD Timeout	
2907	Invalid VFD retries	
2908	Invalid VFD serial mode	
2909	Invalid VFD Device type	
2910	Invalid VFD Modbus context create error	
2911	Invalid VFD Modbus connection error	
2912	Invalid VFD Modbus serial mode error	
2913	Invalid Generic Read Write error	
2914	ABB Handshake error	
2915	ABB Write1 seed state error	
2916	ABB Write2 seed state error	
2917	ABB Write3 seed state error	
2918	ABB Write4 seed state error	
2919	ABB key1 read error	
2920	ABB key2 read error	
2921	ABB key3 read error	
2922	ABB key4 read error	
2923	ABB key1 match error	
2924	ABB key2 match error	
2925	ABB key3 match error	

Event Id	Description	Comments
2926	ABB key4 match error	
2927 - 2999	Reserved	
Accumulators (3000 – 3099)		
3000	Accumulator thread create error	
3001	Invalid DI Channel	
3002	Invalid user interval	
3003	Invalid user period	
3004	Invalid PPM	
3005 - 3099	Reserved	
Data Adaptor (3100 – 3699)		
31000	DA thread create error	
3101	ZMQ Context Error	
3102	ZMQ Socket exit semaphore error	
3103	ZMQ Socket create error	
3104	ZMQ Request/Reply socket bind error	
3105	ZMQ publisher socket bind error	
3106	ZMQ high speed socket bind error	
3107	Jason Message Error	
3108	Jason First Token Error	
3109	Invalid Unique request ID	
3110	Invalid Accumulator value array	
3111	Invalid Accumulator index	
3112	Invalid Analog value array	
3113	Invalid Analog index	
3114 - 3699	Reserved	

2 Accessories

Accessory Item	Description
Power Supply	Double insulated, ATEX-certified 12 V, 60 W power supply
Wi-Fi Antenna & Cable Assembly	LS Research 001-0001 center-fed dipole antenna and LS Research 080-0001 U.FL to Reverse Polarity SMA Connector cable
	Ethertronics Prestta 1000423 and Johnson Emerson U.FL to U.FL coaxial cable 415-0088-150
	Taoglas FXP831.07.0100C flexible dipole antenna with integrated cable and U.FL connector
mPCIe based GSM & CDMA Modem	HE910 Telit make GSM Modems
	DE910 Telit make CDMA Modems
RS-485 Cable	Double shielded twisted pair 18 AWG Ethernet cable
RS-232 Cable	GE Lufkin defined
Debug Port Cable	GE part # 116T7500P0001
Battery	Rayovac make BR2032-BA– BAT,LIT,3V,195MAH,COIN,CELL,-40C,+85C,OPER
LCD Display	AUO Make p/n# G065VN01 V2
Keypad	GE Lufkin defined
LCD & KEYPAD Cables	GE Lufkin defined

The CCGA can operate with the following accessories: MicroSD card, RTC battery, SIM card, and cellular modem. The following are replacement/installation details for these accessories. Refer to the figure, *CCGA Internal Components* for the location of these accessories.

2.1 MicroSD Card

A MicroSD can used by trained GE engineers for software upgrades at the Linux prompt. There is a dip switch internal to the CCGA that must be moved from its default position prior to cycling power on the CCGA.

➢ **To use the MicroSD Card**

1. Power down the CCGA Module.
2. Insert MicroSD card on the microSD card interface on Top.
3. Open the Top black cover.
4. Slide the DIP switch S2 from off to ON to boot from MicroSD card and upgrade the firmware (by default S2 switch position is kept in off position).
5. Power up the board to upgrade firmware.
6. After the upgrade, Power off the CCGA module.
7. Slide the S2 DIP switch back to its off position.
8. Remove the MicroSD card and place the top black cover.
9. Power ON the CCGA module.

2.2 RTC Battery

The CCGA module is shipped with a battery pre-installed (as displayed in the figure, *CCGA Internal Components*). The battery holder is located below the Supercap, and can be replaced by opening the top cover.

➢ **To replace the battery**

1. Power OFF the CCGA Module.
2. Wait for 1 minute.
3. Open the top cover by loosening the four screws on the edges.
4. Use a small flat-head screw driver to gently pry out the old battery.
5. Insert the new battery.



Caution

Replace battery only with Rayovac BR2032 or GEIP part IC690ACC001B or later. Use of another battery may present a risk of fire or explosion.

Battery may explode if mistreated. Do not recharge, disassemble, heat above 100 °C (212 °F) or incinerate.

2.3 SIM Card

➢ **To install a SIM card**

The SIM card holder is located near the SOM board as displayed in the figure, *CCGA Internal Components*.

1. Slide to unlock and open the SIM card holder.
2. Insert the SIM card into the card holder.
3. slide the holder back to lock it into position.

2.4 Cellular Modem

➢ **To install a cellular modem**

The location of the cellular modem and mounting nuts are clearly displayed in the figure, *CCGA Internal Components*.

1. Insert the Modem into the mPCIe connector.
2. Securely mount it using two screws, inserted into the PEM nuts on the modem edges. Use 2.5 X 4 mm Torque head screws with Loctite.

3 Agency Certifications and Standards

Description	Agency Marking	Comments
N.A. Safety for Programmable Controller for use in Hazardous locations Class 1 Division 2 Groups ABCD Class 1 Zone 2 Gas Group IIC	c UL us LISTED	Certification by Underwriter's Laboratories to UL 61010-1; UL61010-2-201, CSA C22.2 No. 61010-2-201; ISA 12.12.01, CSA C22.2 No 213-M1987, UL 60079-0, UL 60079-11, UL 60079-15, CSA-C22.2 No. 60079-0:11, CSA-C22.2 No. 60079-11:11 and CAN/CSA-C22.2 No. 60079-15:12.
Explosive Atmospheres Directive European Safety for Hazardous Areas Equipment Group II, Category 3, Gas Groups IIC	Ex	Self-declaration in accordance with European Directives. Refer to Declaration of Conformity for applicable standards
European Radio Telecommunications & Terminal Equipment Directive (RTTE)	CE	Self-declaration in accordance with European Directives. Refer to Declaration of Conformity for applicable standard

3.1 Hazardous Locations (HazLoc)

This equipment is suitable for use in Class I Division 2 Groups ABCD, Class I Zone 2, and ATEX Zone 2 locations.



Warning

Refer to *CCGA Communication Gateway Instructions for Safe Use (GFK-2953)* for requirements.

3.2 Federal Communications Commission (FCC)

The TIWI or TIWI-BLE Module is labeled with its own FCC ID and IC Certification Number. The FCC ID and IC certification numbers are not visible when the module is installed inside another device, as such the end device into which the module is installed must display a label referring to the enclosed module. The final end product must be labeled by GE Lufkin in a visible area with the following:

Contains Transmitter Module FCC ID: TFB-TIWI1-01
Contains Transmitter Module IC: 5969A-TIWI101

or

Contains FCC ID: TFB- TIWI1-01
Contains IC: 5969A-TIWI101

GE Internal

FCC related info — This device is granted for use in Mobile only configurations in which the antennas used for this transmitter must be installed to provide a separation distance of at least 20 cm from all person and not be co-located with any other transmitters except in accordance with FCC and Industry Canada multi-transmitter product procedures.

This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.



Caution

Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

4 Release History

Date	Firmware Revision	Hardware Revision	Notes
May, 2015	n/a	A	Initial release of the CCGA

5 Replacement and Spares

Replacement parts may contain static-sensitive components. Therefore, GE ships replacement parts in anti-static bags. When handling electronics, make sure to store them in anti-static bags or boxes and use a grounding strap (per the following Caution criteria).



Caution

To prevent component damage caused by static electricity, treat all boards with static-sensitive handling techniques. Wear a wrist grounding strap when handling boards or components, but only after boards or components have been removed from potentially energized equipment and are at a normally grounded workstation.



Warning

To prevent personal injury or damage to equipment, follow all GE safety procedures, Lockout Tagout (LOTO), and site safety procedures as indicated by Employee Health and Safety (EHS) guidelines.



Warning

This equipment contains a potential hazard of electric shock, burn, or death. Only personnel who are adequately trained and thoroughly familiar with the equipment and the instructions should install, operate, or maintain this equipment.

5.1 Replacement Procedures

The failed device should be returned to GE for repair. Do not attempt to repair it on site.

System troubleshooting should be at the module level. The failed module should be removed and replaced with a known good spare.



Warning

To prevent electric shock, turn off power to the equipment, then test to verify that no power exists in the board before touching it or any connected circuits.



Caution

To prevent equipment damage, do not remove, insert, or adjust board connections while power is applied to the equipment.

5.1.1 CCGA Module Replacement

➢ **To replace the CCGA**

1. Lock Out Tag Out (LOTO) the equipment to isolate power sources.
2. Disconnect the incoming power plug.
3. Disconnect the Ethernet cables.
4. Remove the screws holding the CCGA in place.
5. Install the new CCGA by reversing steps 4 through 1.
6. Use the ToolboxST application to configure and download to the new CCGA. Refer to *GEI-100848* for more information.

5.2 Renewals and Spares

Renewals and spares (or those not under warranty) should be ordered by contacting the nearest GE Sales or Service Office, or an authorized GE Sales Representative.

Prior to ordering a replacement part:

- Determine if the part is under warranty
- Identify the part

Control system electronics are identified by a 12-character alphanumeric part number and a single revision letter. When ordering, be sure to include the complete part number and revision letter. All digits are important when ordering or replacing any device. The factory may substitute newer versions based on availability and design enhancements, however, GE ensures backward compatibility of replacements.

CCGA part number

